

Problem Set 4

Due on Tuesday August 17 at 11:59 PM (if you want feedback)

1. Let

$$f(t, x) = e^{-t\sqrt{x}} \ln(1 + x^2).$$

Assume that x is a differentiable function of t satisfying $x = 4$ and $\frac{dx}{dt} = -2$ when $t = 3$. Find the value of the total time derivative

$$\frac{d}{dt} f(t, x(t))$$

when $t = 3$.

2. Assume that $z = x^3 + y^2$ and that x and y are smooth functions of t . Assume further that when $t = 0$, we have $x = 1$, $y = 2$, $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 0$, $\frac{d^2x}{dt^2} = -1$ and $\frac{d^2y}{dt^2} = 3$. Find the value of $\frac{d^2z}{dt^2}$ when $t = 0$.
3. Write an equivalent double integral with the order of integration reversed.

(a) $\int_0^1 \int_y^{\sqrt{y}} f(x, y) dx dy$

(b) $\int_0^2 \int_1^{e^{2x}} f(x, y) dy dx$

(c) $\int_0^2 \int_{2x}^4 f(x, y) dy dx$

(d) $\int_1^3 \int_0^{\ln x} f(x, y) dy dx$

4. Evaluate each of the following.

(a) $\int_0^1 \int_0^{x^3} 4x^2 y dy dx$

(b) $\int_0^1 \int_x^1 3x \sqrt{x^2 + y^2} dy dx$

5. Let R be the triangular region in the $x - y$ plane with vertices $(0, 0)$, $(1, 1)$, and $(2, 0)$. Evaluate

$$\int \int_R xy dA.$$

6. Let R be the annular region in the $x - y$ plane that lies between the two circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. Evaluate

$$\int \int_R (x^2 + y^2)^{\frac{3}{2}} dA.$$

7. Consider the Black-Scholes framework discussed in Lecture 6. Let V be a European-style derivative security that pays the square of the stock price at time T , i.e. the holder of the option receives the amount

$$V_T = (S_T)^2$$

at time T . The time-0 price of this security is given by

$$V_0 = e^{-rT} \int_{-\infty}^{\infty} e^{2x} \psi(x) dx,$$

where ψ is the density function for a normal random variable with mean

$$\mu = \ln S_0 + \left(r - \frac{1}{2} \sigma^2 \right) T,$$

and variance

$$\nu^2 = \sigma^2 T.$$

- (a) Obtain a formula for V_0 by calculating the above integral. (Try to “clean up” the formula as much as possible.) This computation should be quite a bit simpler than the put price that I calculated in lecture.
- (b) The answer for (a) is a function of S_0, T, r, σ , i.e.

$$V_0 = F(S_0, T, r, \sigma).$$

Calculate the partial derivative

$$\frac{\partial F}{\partial S_0}(S_0, T, r, \sigma).$$

This quantity is known as Δ_0 . It represents the number of shares of stock needed at time 0 to replicate (i.e., synthetically create) the option.